

# SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem  
Bounded source-aligned checkpoint — 19 July 2026

**Authority.** Corrected French lines 2733–2750; original printed pp. 91–92; physical source-PDF pp. 80–81; recomposed running pp. 72–73. The printed-page-92 marker occurs at line 2743, immediately before the definitions in (2.4). Line 2751 begins the next equivalence argument and is excluded; it is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

We henceforth assume that  $X$  is affine, Noetherian, and regular, and that  $\mathcal{F} = \widetilde{M}$ , where  $M$  is a finitely generated  $A$ -module, necessarily of finite projective dimension. We shall proceed in several steps. First, we find a condition (d), equivalent to (a), and prove that it is also equivalent to (c). Then, using the spectral sequence of the preceding subsection, we prove  $(d) \Rightarrow (b)$ . It remains to prove  $(iii) \Rightarrow (ii)$ ; indeed,  $(i) \Leftrightarrow (ii) \Rightarrow (iii)$  follows immediately from  $(a) \Leftrightarrow (c) \Rightarrow (b)$ .

Let  $x \in U$ . By hypothesis,  $\mathcal{O}_{X,x}$  is a regular local ring. If  $D$  denotes the dualizing functor for  $\mathcal{O}_{X,x}$ , it follows from (V, 2.1) that

$$D H^{i-c(x)}(\mathcal{F}_x) \simeq \text{Ext}_{\mathcal{O}_{X,x}}^{d(x)-i}(\mathcal{F}_x, \mathcal{O}_{X,x}),$$

where

$$d(x) = \dim \mathcal{O}_{X,x} + c(x) = \dim \mathcal{O}_{X,x} + \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \quad (2.2)$$

Since  $X$  is Noetherian and  $\mathcal{F}$  is coherent,

$$D H^{i-c(x)}(\mathcal{F}_x) \simeq (\mathcal{E}xt_{\mathcal{O}_X}^{d(x)-i}(\mathcal{F}, \mathcal{O}_X))_x. \quad (2.3)$$

Moreover, a module vanishes if and only if its dual vanishes (cf. the editor's note (4) on p. 54). For every  $q \in \mathbb{Z}$ , set

$$\begin{cases} S_q = \text{Supp } \mathcal{E}xt_{\mathcal{O}_X}^q(\mathcal{F}, \mathcal{O}_X), \\ S'_q = S_q \cap U, \quad (U = X - Y), \\ Z_q = \overline{S'_q} \cap Y. \end{cases} \quad (2.4)$$